

Data Analysis Document

HealthFirst Care — Appointment, Feedback & Resource Data

This report summarizes the cleaning, manipulation, and trend analysis performed on HealthFirst Care's appointment, patient-feedback, and resource-utilization data. The full cleaned dataset, lookup-based combined table, pivot summaries, and native charts are provided in the companion workbook HealthFirst_Data_Analysis.xlsx.

1. Data Cleaning Summary

Dataset	Raw Rows	Cleaned Rows	Duplicates Removed	Key Missing-Value Fixes
Appointment	200	200	0	11 rows missing an ID (and usually Time/Status too) → "N/A"; 2 missing dates, 9 missing times, 9 missing statuses → "N/A"
Feedback	216 (18 blank rows)	198	18 blank rows removed; 0 true duplicates	6 missing scores → average (6); 3 missing wait times → average (42 min); 7 missing departments, 2 missing comments → "N/A"
Resource	213	200	13 exact duplicate rows	3 missing IDs, 2 missing departments, 4 missing resource types → "N/A"; no missing usage hours

All dates were standardized to MM/DD/YYYY (the source data mixed ISO YYYY-MM-DD with a 2-digit-year DD/MM/YY format, identified by day values above 12) and all times to H:MM AM/PM. Cells that could not be recovered are flagged "N/A" rather than guessed. Full details and cell-level highlighting are in the workbook's ReadMe and Cleaned_* sheets.

Caveat: rows with a missing original ID now share the literal placeholder "N/A" as their AppointmentID/ResourceID. This satisfies the lab's placeholder instruction but means those specific rows no longer have a unique key — they are flagged for follow-up rather than silently treated as fully reliable records.

2. Data Manipulation Summary

Combined_Feedback (workbook) links each feedback record to its patient's appointment using INDEX/MATCH on Patient ID, pulling Appointment Date, Time, and Department, and adds a Satisfaction Level column via =IF(Score>=4,"High","Low").

Limitation to note: several patients have more than one appointment on file. A single-key lookup (INDEX/MATCH, like VLOOKUP) can only return one match, so it returns each patient's first appointment row. This is disclosed in the workbook rather than presented as a complete one-to-many join.

Pivot-style summaries were built for appointments (by department and hour), feedback (satisfaction by department), and resources (average utilization by department) — see Section 3 below and the workbook's Pivot_* sheets, each with a native Excel chart attached and filter arrows enabled on every table.

3. Key Findings & Trends

3.1 Appointment Trends

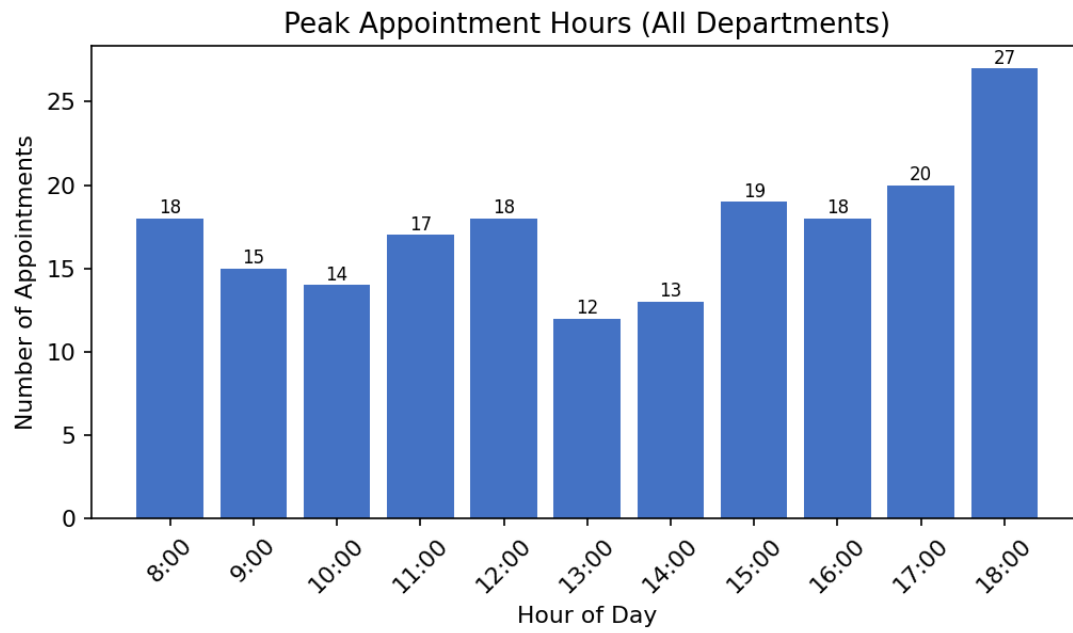


Figure 1 — Appointments by hour of day (all departments).

6:00 PM is the single busiest hour (27 appointments), noticeably above every other slot; 5:00 PM (20) and 3:00 PM (19) are the next busiest. Midday (1:00–2:00 PM) is comparatively quiet (12–13 appointments), suggesting a lunch-hour dip in bookings. Appointment outcomes are also a concern: only 56 of 200 appointments (28%) are marked Completed, while No Show, Rescheduled, and Cancelled together account for 135 appointments (67.5%) — evidence of the same scheduling friction seen in earlier labs, now confirmed at a larger sample size.

3.2 Patient Satisfaction Analysis

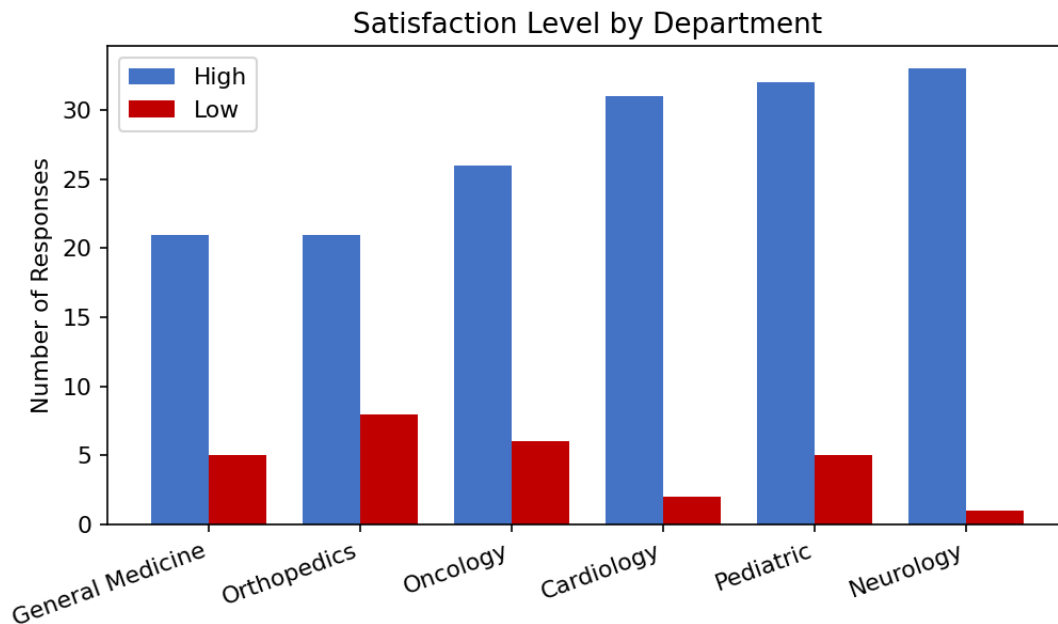


Figure 2 — Satisfaction level ($High \geq 4$, $Low < 4$) by department.

Orthopedics has the highest share of Low-satisfaction feedback (27.6% of its responses), followed by General Medicine (19.2%) and Oncology (18.8%). Neurology (2.9% Low) and Cardiology (6.1% Low) perform best. This ranking is consistent with average feedback scores: Cardiology scores highest (7.03/9) and General Medicine lowest (5.38/9). Among Low-satisfaction responses, "Long wait times" (15 mentions) and "Needs improvement" (12 mentions) are the most common comments — directly consistent with the appointment data's late-afternoon peak and high non-completion rate above.

3.3 Resource Utilization

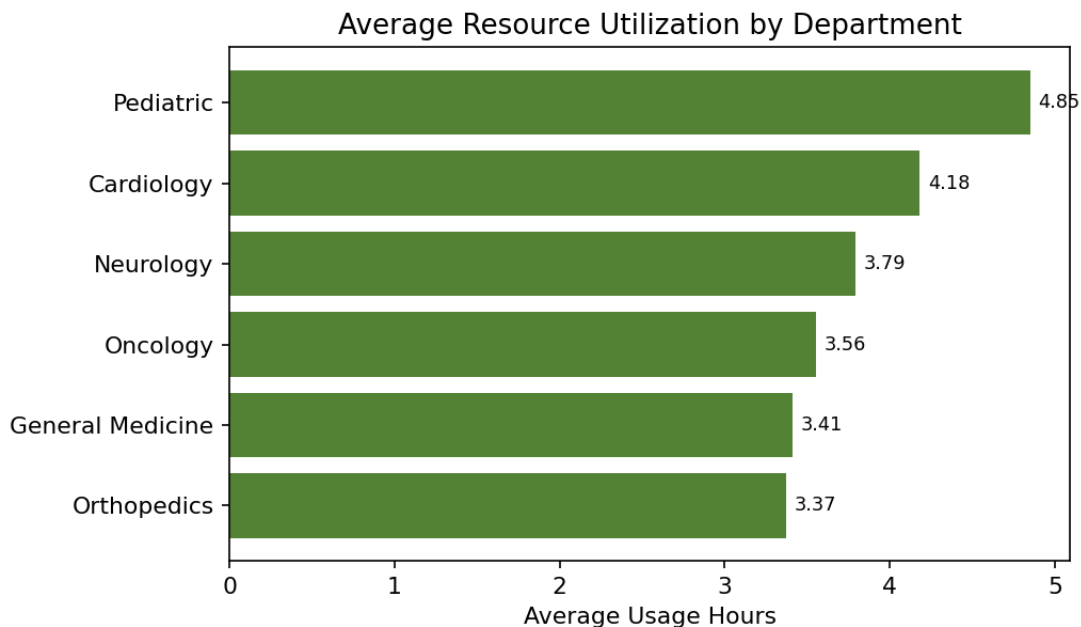


Figure 3 — Average resource usage hours by department.

Pediatric has the highest average resource utilization (4.85 hours) and Orthopedics the lowest (3.37 hours) — yet Orthopedics also has the highest Low-satisfaction rate, suggesting its dissatisfaction is not simply a matter of resource load. Looking at availability status instead of usage hours, General Medicine has the most resources logged "Under Maintenance" (7 of 39 records, 18%) and Pediatric the most "Unavailable" (7 of 40, also tied for highest alongside Oncology's 6 of 36) — these two patterns point to different bottlenecks: General Medicine is more affected by equipment/room downtime, while Pediatric and Oncology are more affected by staff/doctor unavailability.

4. Summary of Recommendations

- Investigate the 6:00 PM scheduling spike specifically — it is the clearest single driver of peak-time load and a natural first target for the automated conflict-detection scheduling improvement already scoped in the BRD (FR1).
- Prioritize Orthopedics and General Medicine for the patient-communication and wait-time improvements (FR2, NFR1), since both combine below-average satisfaction with above-average Low-satisfaction rates.
- Address General Medicine's equipment/room maintenance backlog and Pediatric/Oncology's staff availability gaps as two distinct resource problems, since the data indicates they are not the same bottleneck.
- Treat the 67.5% non-completion rate (No Show + Rescheduled + Cancelled) as the single most consequential appointment metric to track going forward, given how directly it recurs across every dataset reviewed in this analysis.

Note on evidence: figures above are computed directly from the provided (larger, 200-row-scale) datasets and are more statistically meaningful than the smaller earlier samples, but they remain a single data pull for this exercise rather than a full operational extract.

5. Attachments

- HealthFirst_Data_Analysis.xlsx — the full workbook: Raw_* and Cleaned_* sheets for all three datasets, Combined_Feedback with live INDEX/MATCH and IF formulas, and Pivot_Appointments / Pivot_Feedback / Pivot_Resource with native Excel charts (also embedded above).